Monetary Policy in Practice

CHAPTER 15 · PRINCIPLES OF MACROECONOMICS

Output gaps, the Taylor rule, and inflation targeting

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Learning Objectives

- Measure output gaps and evaluate policy responses to demand and supply shocks.
- Calculate and interpret a standard Taylor-rule benchmark.
- Compare a policy-rate rule with an inflation-targeting framework and explain their limits.

Choosing a Policy Response

- The Fed pursues maximum employment and price stability. Its longer-run inflation goal is 2% PCE inflation.
- Compare actual output with sustainable potential output; evaluate inflation and its expected path.
- A demand shortfall and a supply disruption can produce different policy tradeoffs.

Federal Reserve Board: policy goals and transmission

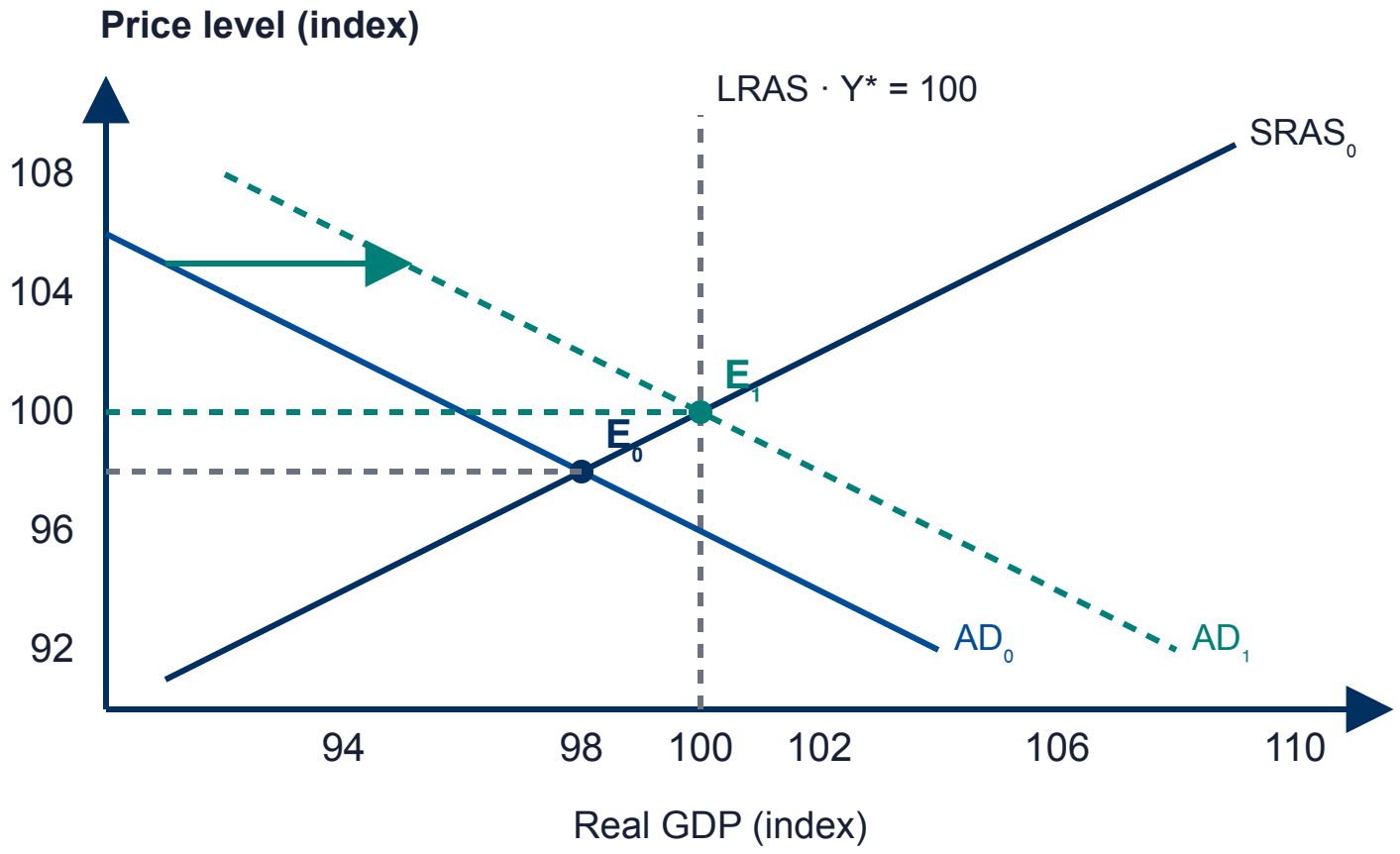
Measuring the Output Gap

Hypothetical case	Actual real GDP	Potential real GDP	Output gap
Demand shortfall	980	1,000	Calculate
Demand pressure	1,020	1,000	Calculate

- Output gap = 100 × (actual GDP − potential GDP) ÷ potential GDP.
- Shortfall: 100 × (980 − 1,000) ÷ 1,000 = −2%.
- Pressure: 100 × (1,020 − 1,000) ÷ 1,000 = +2%.

All GDP inputs are original illustrations in the same real units; potential GDP is estimated.

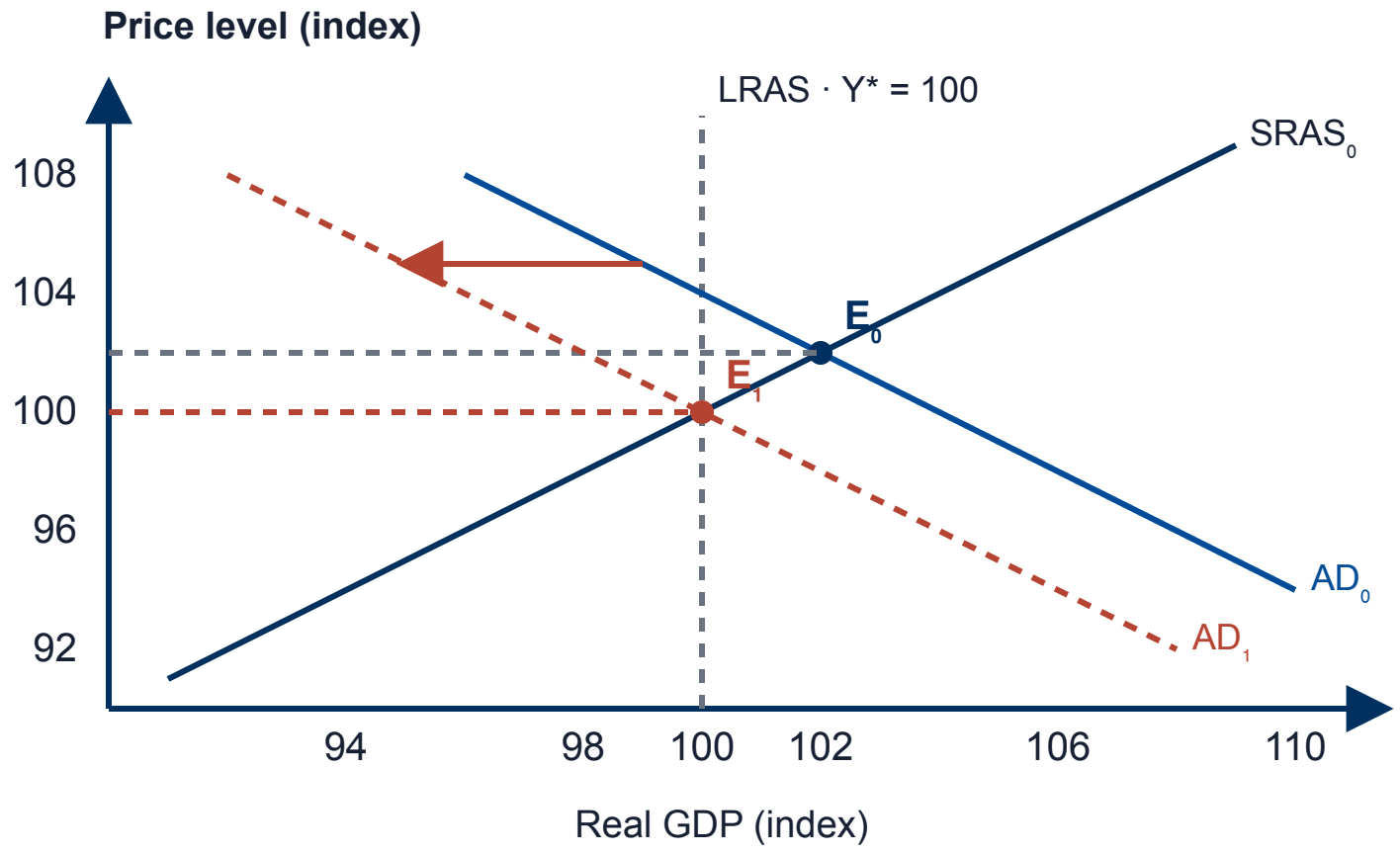
Responding to a Negative Output Gap



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. A negative demand shock leaves output below potential.
2. If inflation pressure is subdued, easing can support C and I.
3. AD moves right toward the fixed potential-output line.
4. Output rises toward potential; the price level rises relative to the weak-demand outcome.

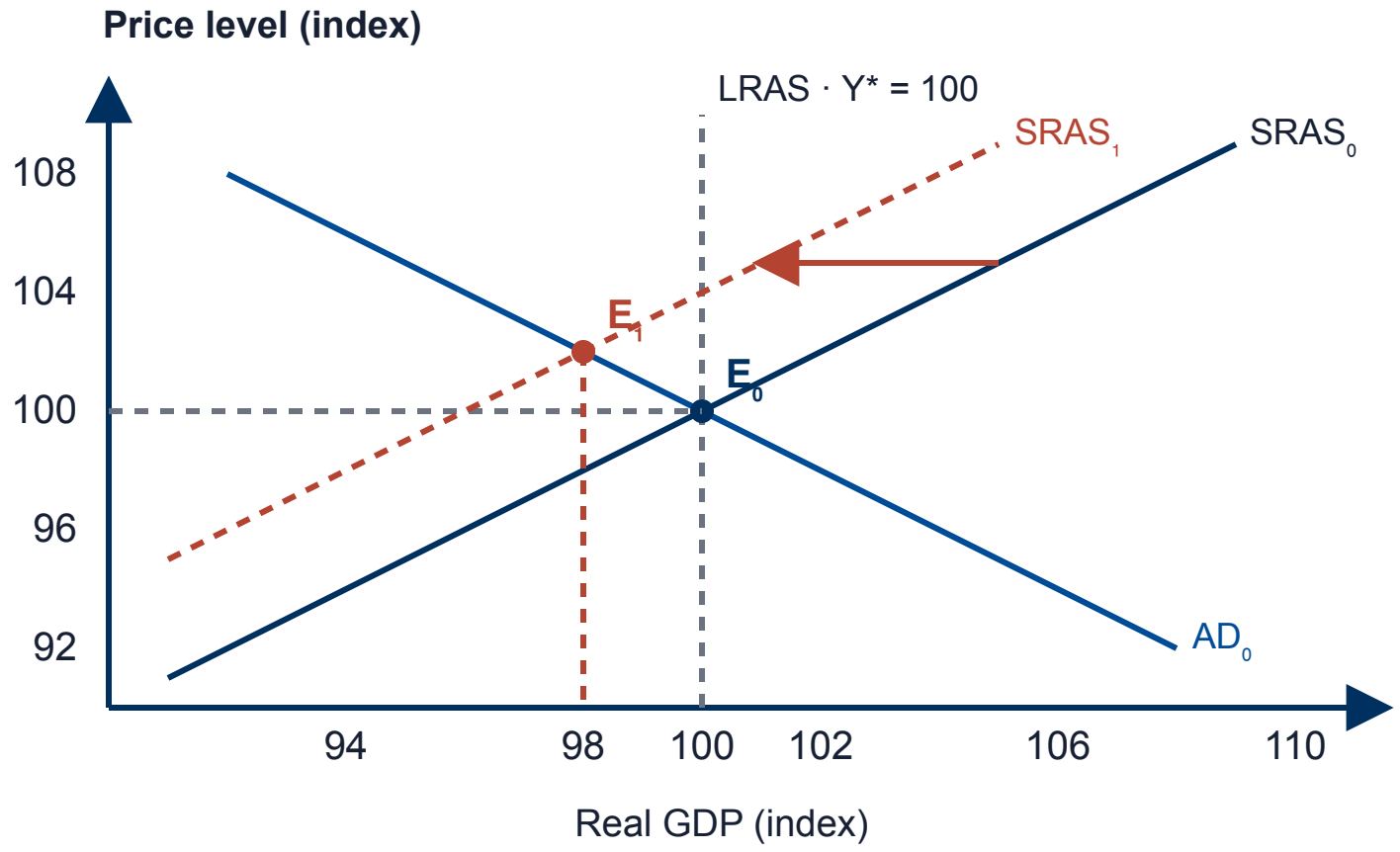
Responding to a Positive Output Gap



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. Strong aggregate demand puts output above sustainable potential.
2. If inflation pressure is excessive, tightening can restrain C and I.
3. AD moves left toward the fixed potential-output line.
4. Output pressure eases, with a lower price level than without tightening.

When Output and Inflation Disagree



Original illustrative model; SRAS and potential output stay fixed during the AD policy comparison.

1. An adverse supply shock shifts SRAS left/up.
2. Output falls below potential while the price level rises.
3. Easing AD can support output but adds price pressure; tightening AD can reduce price pressure but deepen the output loss.

A Standard Taylor Rule

$$i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5g$$

- i: nominal policy-rate benchmark; r*: neutral real rate.
- π: inflation; π*: inflation goal; g: output gap in percent.
- Higher inflation relative to the goal or a larger output gap raises the benchmark, holding other inputs fixed.

Illustrative 1993-style rule. Rates and gaps enter as percentage numbers, such as 4 and −2.

Federal Reserve Board: policy rules

Calculating a Policy-Rate Benchmark

Fixed illustrative input	Value
Neutral real rate r^*	2%
Inflation goal π^*	2%
Current inflation π	4%
Output gap g	−2%

- Baseline term: $r^* + \pi = 2 + 4 = 6$.
- Inflation response: $0.5 \times (4 - 2) = +1$.
- Output response: $0.5 \times (-2) = -1$.
- Benchmark: $i = 6 + 1 - 1 = 6\%$.

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Why the Rule Responds to Inflation

> **$\Delta i = 1.5 \times \Delta \pi$ (holding r^*, π^*, and g fixed)**

- One extra percentage point of inflation adds 1 point through π and 0.5 point through the inflation-gap term.
- The nominal rate rises by 1.5 percentage points.
- If expected inflation rises by 1 point too, the approximate real rate rises by 0.5 point.

Federal Reserve Board: policy rules

Inflation Targeting

- An inflation-targeting framework states an inflation objective and uses forecasts, communication, and policy actions to pursue it over time.
- For the Fed, the longer-run objective is 2% PCE inflation, alongside maximum employment.
- Policy can respond to expected inflation and employment conditions before inflation reaches the goal.

Federal Reserve Board: policy goals and transmission

The September 2024 Decision

Case E · September 18, 2024	Policy fact
Target range change	Cut by 0.50 percentage point
New target range	4.75–5.00%
Statement context	Inflation still somewhat elevated; greater confidence in progress toward 2%

- The committee considered progress on inflation and risks to both employment and inflation.
- A cut before inflation reaches 2% can reflect a forward-looking judgment. The statement does not prove a specific Taylor-rule calculation.

FOMC, September 18, 2024

Taylor Rule and Inflation Targeting

Comparison	Taylor-rule benchmark	Inflation-targeting framework
Main question	What policy rate do these inputs imply?	What outcome should policy pursue over time?
Central inputs	Inflation, inflation gap, output gap, r*	Inflation objective, forecasts, risks, broader mandate
Type of guidance	A numerical instrument guide	An objective and decision framework
Can they coexist?	Yes: a rule can inform the rate decision.	Yes: the target can anchor the rule.

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Why Policymakers Use Judgment

- Potential output and the neutral real rate are estimates that can change.
- Inflation measures and forecasts can disagree; effects of policy arrive with lags.
- The effective lower bound can limit conventional rate cuts. A formula can prescribe an infeasible rate.

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Try the Rule

Fixed input	Value
r* and π*	2% each
Inflation π	3%
Output gap g	−4%

- Calculate the nominal policy-rate benchmark.
- Explain why a negative output gap pushes the benchmark downward.

Explanation. i = 2 + 3 + 0.5(3 − 2) + 0.5(−4) = 3.5%. The output-gap term contributes −2 percentage points.

Write a Conditional Policy Recommendation

Hypothetical economy	Inflation	Output gap	Shock
A	1%	−3%	Demand shortfall
B	5%	−3%	Adverse supply shock

- For each case, recommend a direction and trace the channel through AD.
- Calculate a Taylor benchmark using $r^* = 2\%$, $\pi^* = 2\%$, then explain what the number leaves unresolved.
- Name one piece of evidence that could change your recommendation.

Explanation. A: benchmark 1%; easing can support weak demand if conditions allow. B: benchmark 7%; inflation and output signals conflict, so discuss the tradeoff, expectations, persistence, and other available tools before choosing a response.

Federal Reserve Board: policy rules

Policy Reasoning

- Identify the shock and the output gap before recommending a policy.
- Use a rule to organize the evidence; distinguish a benchmark from a binding decision.
- Explain the expected channel, the tradeoff, and the condition that could change your answer.

Explanation. An output gap describes the economy’s position. A policy rule supplies a numerical cross-check. An inflation objective and mandate guide the decision over time.